SOFTWARE INNOVATION STUDIO · GROUP 18

# ClaimTrace.

*Check what your citations actually say.*

WEB WORKSPACE  OVERLEAF EXTENSION  PDF PARSER  VERIFICATION ENGINE

> *"I want the evidence, not just the conclusion — so I can tell the student exactly what to fix."*
>
> Supervisor persona · translated from user research

## Three checks, in the place you already write.

01 / WHAT IT DOES

01 / AUDIT

### Audit

every reference, every field

Every bibliography entry is checked against OpenAlex, then Crossref. Title, authors, year and venue are compared field by field — and a lookup that failed stays visibly failed.

02 / VERIFY

### Verify

claim vs. what the paper says

A selected sentence is matched to the cited paper. The relevant passages are retrieved and every verdict ships with the passage, its rank and its similarity score.

03 / READ

### Read

the paper as it was written

PDFs become structured Markdown — formulas, tables and figures preserved — and render inside the review pane, next to the manuscript.

## A reference is either checked, or it is visibly not checked.

02 / AUDIT

| REFERENCE | STATE | EVIDENCE |
|---|---|---|
| `bert-exact`<br>Devlin et al., 2019 | ● VERIFIED | All four fields match the OpenAlex record. One lookup; Crossref is never asked.<br>`title · authors · year · venue → MATCH` |
| `bert-year-plus-one`<br>year stated one late | ● METADATA MISMATCH | The record stays eligible inside the one-year tolerance, so the difference is reported field by field.<br>`year → MISMATCH, and the rest of the record is still checked` |
| `dialoguernn-year-minus-five`<br>year stated five early | ● NOT FOUND | A five-year gap rejects every record both providers return, so the report carries `field_checks: []` rather than a fabricated mismatch.<br>`no acceptable record → nothing to compare` |
| `provider-500`<br>one provider errors | ● LOOKUP FAILED | Crossref's search is negative and OpenAlex cannot answer. The reference stays visibly unchecked — never reported as "not found".<br>`an unread reference is not a missing reference` |

FIVE STATES ● VERIFIED ● METADATA MISMATCH ● NEEDS REVIEW ● NOT FOUND ● LOOKUP FAILED

"Lookup failed" is a state, not a missing reference.

## The verdict is only the headline. The passage is the argument.

03 / VERIFY

CLAIM 01

"The paper reports that the method reduced carbon emissions by forty percent."

● NOT FOUND  The source was read and does not contain this claim. That is different from failing to read it.

CLAIM 02

"Self-attention relates different positions of a single sequence."

> "Self-attention, sometimes called intra-attention, is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence."

● SUPPORT  Retrieved from the cited paper · quoted verbatim.

## One process, two outbound boundaries.

04 / HOW IT WORKS

CLIENT
React web workspace

CLIENT
Overleaf extension

→

BACKEND
FastAPI · /api/*

→

IN-PROCESS
Engine + Parser

→

OUTBOUND
OpenAlex · Crossref
your own model key

A failed lookup is never reported as "not found". In Verify, only `COMPARED` carries a verdict.

## Measured, not claimed.

05 / EVIDENCE

**1,130**
AUTOMATED TESTS
ACROSS FIVE SUITES

**34/34**
CONTROLLED AUDIT CASES
EXACT · MACRO-F1 1.00

**12/12**
LIVE VERIFY CASES
DEEPSEEK · 12 CALLS

**0.055 s**
PER AUDIT REQUEST
LINEAR SCALING

Audit measured on 31 .bib and 3 PDF fixtures with pinned provider responses (generated 2026-10-08): state accuracy 1.00, every field check exact. Verify measured on six manually labelled claims in two modes; labels are fixture labels, not independent expert annotation. Test count at local main e171fb3.